INCH-POUND
MIL-DTL-17/132C
w/AMENDMENT 2
22 August 2025
SUPERSEDING
MIL-DTL-17/132C
w/AMENDMENT 1
9 January 2016

MILITARY SPECIFICATION SHEET

CABLES, RADIO FREQUENCY, FLEXIBLE, COAXIAL, LOW NOISE, 50 OHMS, M17/132-00001

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17.

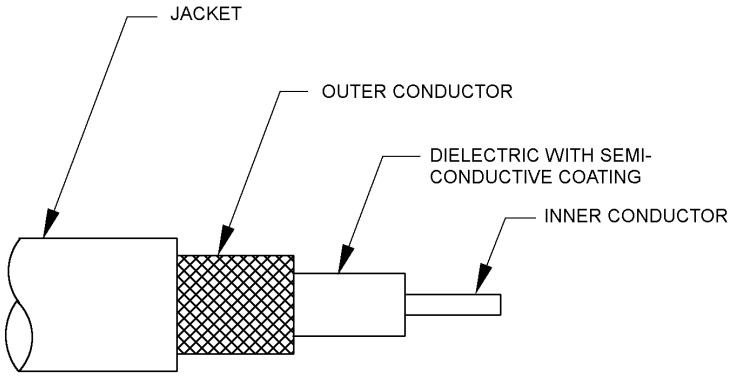


FIGURE 1. Configuration.

TABLE I. Description.

Components	Construction details
Inner conductor	Seven strands of AWG size 38 silver-plated, copper-covered steel wire. Overall diameter: 0.012 ± .001 inch.
Dielectric	Type F-1 with a semiconducting coating. Overall diameter: .036 ± 0.002 inch.
Outer conductor	Silver-coated, AWG size 38 copper strands. Overall diameter: 0.056 inch, maximum. Carriers: 16 Ends: 3 Picks/inch: 25 ± 10 percent.
Jacket	Type IX. Overall diameter: .073 ± .004 inch.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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ENGINEERING INFORMATION:

Continuous working voltage: 1,500 V rms, maximum.
Operating frequency: 1 GHz, maximum.
Velocity of propagation: 68 percent, nominal.
Operating temperature range: -40°C to +200°C.
Inner conductor properties:
DC resistance (maximum at 20°C): 24.45 ohms per 100 feet.
Elongation: 10 percent, minimum.
Tensile strength: 50 $klb_f/inch^2$, minimum.

Engineering notes: This cable is useful in low noise temperature applications.

REQUIREMENTS:

Dimensions, configuration, and descriptions: See figure 1 and table I.

Environmental and mechanical:
Adhesion of conductors:
Inner conductor to core: 1 pound, minimum; 4 pounds, maximum.
Time delay: Not applicable.
Aging stability: Not applicable.
Stress crack resistance: +230°C ± 5°C for 76 hours.
Outer conductor integrity: Not applicable.
Cold bend: -55° ± 2°C.
Dimensional stability: +200° ± 5°C.
Inner conductor from core: .062 inch, maximum.
Inner conductor from jacket: .125 inch, maximum.
Contamination: Not applicable.
Bendability: Not applicable.
Flammability: Not applicable.
Weight: 1.65 pounds per 100 feet, maximum.

Electrical:

Test frequency: 50 MHz to 1 GHz.
Continuity: Applicable
Spark test: 2,000 Vrms, +25 percent, -0 percent.
Voltage withstanding: 1,500 Vrms, minimum.
Insulation resistance: Not applicable.
Corona extinction voltage: 1,000 V rms, minimum.
Characteristic impedance: 50 ohms ± 2 ohms.
Attenuation: 29dB/100 feet maximum at 400MHz.
Capacitance: 32 pF per foot, maximum.
Capacitance stability: Not applicable.
Capacitance unbalance: Not applicable.
Transmission unbalance: Not applicable.
Mechanically induced noise voltage: 25 millivolts, peak to peak maximum.
Time delay: Not applicable.

Verification: See table III.

Marking: Applicable

Marking Durability: 125 cycles (250 strokes), 1 pound.

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Workmanship: Applicable.

Part or Identifying Number (PIN): M17/132-00001.

Supersession data: See table II.

TABLE II. Cross reference of PIN.

PIN	Superseded PIN	Superseded type designation
M17/132-00001	M17/132-RG404	RG-404/U

TABLE III. Qualification and Group A, B, and C.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Qual- # of Specimens	Conf - # of Specimens	Conformance inspection
I	In-process inspection:		4.5	Entire sample		
	Continuity	3.7.1	4.8.2	Entire sample		
	Spark test	3.7.2	4.8.3	Entire sample		
	Voltage withstanding	3.7.3	4.8.4	Entire sample		
	Insulation resistance	3.7.4	4.8.5	Entire Sample		
	Visual and mechanical inspection	3.6	4.8.1	Entire sample	Per Table VIII	A
	Diameter measurements.	3.6.1	4.8.1.1	Entire sample	Per Table VIII	A
	Out-of-roundness	N/A				
	Eccentricity of inner conductor	N/A				
	Adhesion of conductors	3.6.4	4.8.1.4	4	Per Table VIII	A
	Marking	3.8	4.8.1	Entire sample	Per Table VIII	A
	Marking Durability	3.8.2	4.8.39	3	Per Table VIII	A
	Workmanship	3.10	4.8.1	Entire sample	Per Table VIII	A
II	Corona extinction voltage	3.7.5	4.8.6	1	Per Table VIII	B
	Characteristic impedance	3.7.6	4.8.7	1	Per Table VIII	A
	RF transmission loss (attenuation)	3.7.7	4.8.8	2	Per Table VIII	A
	Standing wave ratio (return loss)	N/A				
	Capacitance	3.7.9	4.8.10	1	Per Table VIII	B
	Capacitance stability	N/A				

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TABLE III. Qualification and Group A, B, and C – Continued.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Qual- # of Specimens	Conf - # of Specimens	Conformance inspection
II	Capacitance unbalance	N/A				
	Transmission unbalance	N/A				
	Inductance	N/A				
	Mechanically induced noise voltage	3.7.13	4.8.14	1	Per Table VIII	**B**
	Time delay	N/A				
	Aging stability	N/A				
	Stress crack resistance	3.7.16	4.8.17	4	Per Table VIII	C
	Outer conductor integrity	N/A				
	Cold bend	3.7.18	4.8.19	3	Per Table VIII	B
	Dimensional stability	3.7.19	4.8.20	1	Per Table VIII	C
	Contamination	N/A				
	Bendability	N/A				
	Flammability	N/A				
	Flame propagation	N/A				
	Acid gas generation	N/A				
	Halogen content	N/A				
	Immersion	N/A				
	Smoke index	N/A				
	Toxicity index	N/A				
	Durometer hardness	N/A				
	Weathering	N/A				
	Abrasion resistance	N/A				
	Tear strength	N/A				
	Heat distortion	N/A				
	Physicals	N/A				
	Tensile strength and elongation	N/A				
	Weight	3.9	4.8.24	1	Per Table VIII	B

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Referenced documents. This document references MIL-DTL-17.

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project 6145-2025-016)

Review activities:
Army – AT, CR4, MI
Navy – AS, MC, OS, SH
Air Force – 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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